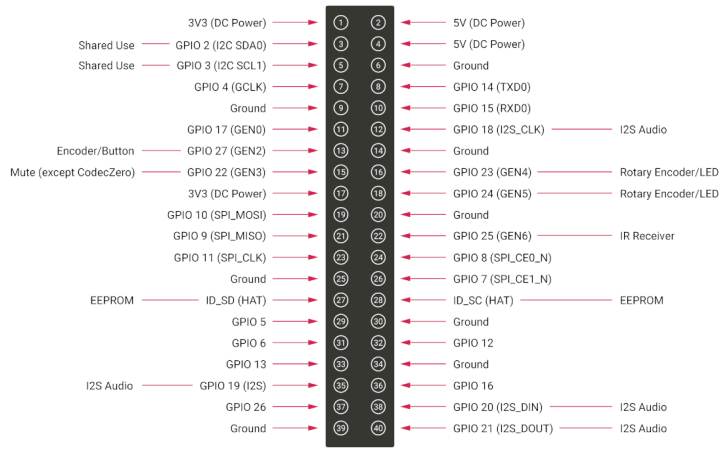
Raspberry Pi audio board hardware information

Source: https://github.com/raspberrypi/documentation/blob/7376533ab86ea07acc77ef6a9f3c3e57fcff833d/documentation/asciidoc/accessories/audio/hardware-info.adoc (CC BY-SA 4.0; adapted for this benchmark)
Image source: https://raw.githubusercontent.com/raspberrypi/documentation/7376533ab86ea07acc77ef6a9f3c3e57fcff833d/documentation/asciidoc/accessories/audio/images/all_audio_boards_gpio_pinouts.png (CC BY-SA 4.0)



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Hardware information

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PCB screws are all M2.5.

PCB standoffs (for case) are 5 mm male/female.

PCB standoffs (for Raspberry Pi to audio boards) are 9 mm female/female.

PCB standoffs (for XLR to DAC PRO) are 8 mm female/male.

PCB standoffs (for the official Raspberry Pi 7-inch display) are 5 mm male/female.

The rotary encoders we have used and tested are the Alpha three-pin rotary encoder

RE160F-40E3-20A-24P, the ALPS EC12E2430804 (RS: 729-5848), and the Bourns ECW0JB24-AC0006L (RS: 263-2839).

The barrel connector used for powering the DigiAMP+ is 2.5 mm ID, 5.5 mm OD, 11 mm.

The DigiAMP+ is designed to operate with a 12V to 24V, 3A supply such as the XPPower

VEC65US19 or similar.

The DigiAMP+ uses CamdenBoss two-part connectors. Those fitted to the PCB are

CTBP9350/2AO.

The speaker terminal used on the Codec Zero will accept wires of between 14~26 AWG

(wire of max 1.6 mm in diameter).

GPIO usage

Raspberry Pi audio boards take advantage of a number of pins on the GPIO header in

order to operate successfully. Some of these pins are solely for the use of the board, and

some can be shared with other peripherals, sensors, etc.

The following Raspberry Pi GPIO pins will be used by the audio boards:

All power pins

All ground pins

GPIO 2/3 (I2C)

GPIO 18/19/20/21 (I2S)

If appropriate then the following are also used:

GPIO 22 (DigiAMP+ mute/unmute support)

GPIO 23/24 for rotary encoder (physical volume control) or status LED (Codec Zero)

GPIO 25 for the IR Sensor

GPIO 27 for the rotary encoder push switch/Codec Zero switch

DAC PRO, DAC+, DigiAMP+, Codec Zero

The DAC PRO, DAC+ and DigiAMP+ re-expose the Raspberry Pi signals, allowing additional sensors and peripherals to be added easily. Some signals are for exclusive use (I2S and EEPROM) by some of our boards; others, such as I2C, can be shared across multiple boards.

Saving AlsaMixer settings

To store the AlsaMixer settings, add the following at the command line:

$ sudo alsactl store

You can save the current state to a file, then reload that state at startup.

To save, run the following command, replacing <username> with your username:

$ sudo alsactl store -f /home/<username>/usecase.state

To restore a saved file, run the following command, replacing <username> with your username:

$ sudo alsactl restore -f /home/<username>/usecase.state

MPD-based audio with volume control

To allow Music Player Daemon (MPD)-based audio software to control the audio board's built in volume, you might need to change the file /etc/mpd.conf to support the correct AlsaMixer name.

This can be achieved by ensuring the 'Audio output' section of /etc/mpd.conf has the 'mixercontrol'

line. Below is an example for the Texas Instruments-based boards (DAC

PRO/DAC+/DigiAMP+):

audiooutput {

type "alsa"

name "ALSA Device"

mixercontrol "Digital"

}